

Table of contents

3 次対称群 $G = \mathfrak{S}_3 = \{1, (123), (132), (12), (13), (23)\}$ の乗法表

$1 \mid (123) \mid (132) \mid (12) \mid (13) \mid (23) \mid$

$\begin{array}{cccccccc} \mid & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ \mid & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \end{array} \mid 1 \mid (123) \mid (132) \mid (12) \mid (13) \mid (23) \mid$

$\mid (123) \mid (132) \mid (132) \mid 1 \mid (13) \mid (23) \mid (12) \mid \mid (132) \mid (132) \mid 1 \mid (123) \mid (23) \mid (12) \mid (13) \mid \mid (12) \mid (12) \mid (23) \mid (13) \mid 1 \mid$
 $\mid (132) \mid (123) \mid \mid (13) \mid (13) \mid (12) \mid (23) \mid (123) \mid 1 \mid (132) \mid \mid (23) \mid (23) \mid (13) \mid (12) \mid (132) \mid (123) \mid 1 \mid$

部分群は

$H = \{1, (123), (132)\}$

$K_1 = \{1, (12)\}$

$K_2 = \{1, (13)\}$

$K_3 = \{1, (23)\}$

H に対して, $C(1) = \{1, (123), (132)\}$, $C((12)) = \{(12), (13), (23)\}$

K_1 に対して, $C(1) = \{1, (12)\}$, $C((123)) = \{(123), (13)\}$, $C((132)) = \{(132), (23)\}$

K_2 に対して, $C(1) = \{1, (13)\}$, $C((123)) = \{(123), (23)\}$, $C((132)) = \{(132), (12)\}$

K_3 に対して, $C(1) = \{1, (23)\}$, $C((123)) = \{(123), (12)\}$, $C((132)) = \{(132), (13)\}$

$(12)H = \{(12), (23), (13)\}$

$(13)H = \{(13), (12), (23)\}$

$(23)H = \{(23), (13), (12)\}$